

Mathematics & Statistics
MATH 1B03
Linear Algebra I
Fall 2026



SCIENCE

Instructor Information

Dr. Erin Clements
Email: clemene@mcmaster.ca
Office Hours:
TBA

Course Information

Lectures: See personal timetable in Mosaic

Tutorials: See personal timetable in Mosaic

Course Dates: 09/08/2026 - 12/10/2026

Units: 3.00

Course Delivery Mode: In Person

Course Description: Vector spaces given by solutions to linear systems. Linear independence, dimension. Determinants. Eigenvalues, eigenvectors and diagonalisation. Complex numbers. Three lectures, one tutorial; one term

Prerequisite(s): Grade 12 Calculus and Vectors U or MATH 1F03 Antirequisite(s): MATH 1ZC3 Not open to students registered in an Engineering program.

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

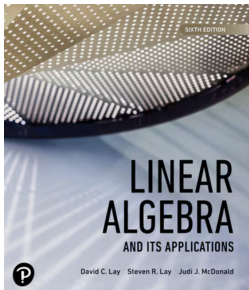
- [Childsmath](#)

Course Learning Outcomes

- Connect computational techniques with theoretical ideas in linear algebra to interpret and justify results involving matrices and vector spaces
- Demonstrate mathematical reasoning and critical thinking in both computational and theoretical contexts
- Perform matrix computations, including solving systems of linear equations using Gauss–Jordan elimination, carrying out matrix arithmetic, computing determinants, and finding eigenvalues and eigenvectors
- Use vector space concepts, including linear independence, span, basis, and dimension, to analyze and describe solution sets and matrix structures

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist



Linear Algebra and its Applications

Authors: D. Lay, S. Lay, and J. McDonald

Publisher: Pearson

Edition: 6th

We will not be using the associated MyLab material; you don't need to purchase

MATLAB software

We will use MATLAB for computer-based computations, as it is widely used for mathematical and scientific computing. Tutorials in [Childsmath](#) explain the basics of using MATLAB. As a McMaster student, you can obtain a version of MATLAB through University Technology Services ([UTS](#)).

Class Format

In Person

There will be two sections of MATH 1B03 offered in the Fall term. These sections will be taught in a coordinated manner, with common tests, assignments, and a common final exam.

Students will attend three hours of lectures per week. In addition, several tutorial sections will be offered, and students are strongly encouraged to attend their assigned tutorial section. Tutorials are designed to supplement lectures by providing additional problem-solving practice, support with computer usage, and assistance with homework.

Course Evaluation

Assessment	Weight
Assignments	10%
Labs	5%
Tests	40%
Final Exam	45%

Course Evaluation Details

Assignments

- There will be a series of online assignments corresponding to lecture topics, assigned regularly throughout the semester. These assignments will be accessed and submitted through Childsmath. Details (e.g., assignments due dates, new assignments postings) will be provided in the Schedule and Homework section of our Avenue page.

Labs

- There will also be a series of online computer labs to be completed in MATLAB, with answers submitted through Childsmath. Each lab has associated tutorials available on Childsmath that should be reviewed before attempting the lab. In addition, in-person computer tutorials are scheduled to support each lab. Details (e.g., lab due dates, new lab

postings) will be provided in the Schedule and Homework section of our Avenue page.

Tests

- There will be two tests, which will be held in person during the evening. Details (e.g., material that will be covered, test procedure, etc.) will be announced on Avenue approximately one week before the test. *Only the standard McMaster calculator, Casio fx991MS or Casio fx991MS PLUS, may be used during tests.*

Final Exam

- There will be a cumulative, 2.5 hour final exam scheduled during the December exam session. This exam will be scheduled by the Registrar and date/time will be announced later in the semester. Details (e.g., material that will be covered) will be announced on Avenue approximately one week before the exam. *Only the standard McMaster calculator, Casio fx991MS or Casio fx991MS PLUS, may be used during the exam.*

Undergraduate Grading Scale

The grades breakdown for McMaster's 12-Point Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Graduate Grading Scale

Graduate Student Grading Scale (Except for MBA and Master of Finance)

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+
A	11	85-89	
A-	10	80-84	
B+	9	77-79	P
B	8	73-76	
B-	7	70-72	
F	0	69 and under	F

MBA and Master of Finance Grading Scale

Grade	Points	Equivalent Percentage	Pass/Fail
A+	12	90-100	P+
A	11	85-89	
A-	10	80-84	
B+	9	75-79	P
B	8	70-74	
B-	7	60-69	
F	0	59 and under	F

Course Schedule

**** Tentative ** Schedule (see schedule in Avenue for updated details)**

Week	Topic
1	Introduction 1.1 Systems of Linear Equations 1.2 Row Reduction and Echelon Forms
2	1.2 Row Reduction and Echelon Forms (continued) <i>Introduction to Matlab</i> 1.3 Vector Equations 1.4 Matrix Equation $Ax = b$
3	1.5 Solution Sets of Linear Equations 1.7 Linear Independence 1.8 Introduction to Linear Transformations
4	1.9 Matrix of a Linear Transformation 1.6 Applications of Linear Systems 2.1 Matrix Operations
5	2.2 The Inverse of a Matrix (two lectures)

Week	Topic
6	2.3 Characterizations of Invertible Matrices (two lectures) 2.4 Partitioned Matrices 3.1 Introduction to Determinants
7	<i>Reading Week</i>
8	3.2 Properties of Determinants 3.3 Cramer's Rule, Volume, and Linear Transformations 4.1 Vector Spaces and Subspaces
9	4.1 Vector Spaces and Subspaces (continued) 4.2 Null Spaces, Column Spaces, and Linear Transformations (two lectures)
10	4.3 Linear Independent Sets and Bases 4.4 Coordinate Systems 4.5 Dimension of a Vector Space
11	5.1 Eigenvectors and Eigenvalues 5.2 The Characteristic Equation Appendix B Introduction to Complex Numbers
12	Complex eigenvalues 5.3 Diagonalization (two lectures)
13	5.4 Eigenvectors and Linear Transformations 6.1 Inner Product, Length, and Orthogonality 6.2 Orthogonal Sets 6.3 Orthogonal Projections
14	6.4 Gram-Schmidt Process Review

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Course Modification

The instructor reserves the right to change the weight of any portion of this marking scheme. For students in good academic standing, other weights might be considered. In either case, the final mark will be computed using this weighting and the new weighting(s). The highest score for a particular student will be their final mark.

At the end of the course the grades may be adjusted but this can only increase your grade and will be done uniformly. We will use the grade equivalence chart published in the Undergraduate Calendar to convert between percentages and letter grades.

Generative AI: Some Use Permitted

Students may use generative AI in this course in accordance with the guidelines outlined for each assessment, and so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the limitations for use for each assessment and to be clear on the expectations for citation and reference and to do so appropriately.

Course Specific Relief for Missed Academic Work Information

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

In Math 1B03:

- If an MSAF is submitted for one **missed assignment**, the weight of the missed assignment will be transferred to the remaining assignments. If an assignment is missed but no MSAF is submitted, then a grade of 0 will be recorded. If more than one assignment is missed, contact the instructor.

- If an MSAF is submitted for one **missed lab**, the weight of the missed lab will be transferred to the remaining labs. If a lab is missed but no MSAF is submitted, then a grade of 0 will be recorded. If more than one lab is missed, contact the instructor.
- If an MSAF is submitted for a **missed test**, the weight of the missed test will be transferred to the final exam. If a test is missed but no MSAF is submitted, then a grade of 0 will be recorded.

Checking Grades & Regrade Policy

Checking Your Grades

Your marks will be recorded on Avenue and Childsmath. It is your responsibility to check that all grades entered into Avenue and Childsmath are recorded properly. All grade concerns and discrepancies must be reported to the instructor before the final exam.

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which

- other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that standards of academic integrity have been upheld** (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the “Code”). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online.**

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students’ access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University’s [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office **normally within 10 working days** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.).

Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.